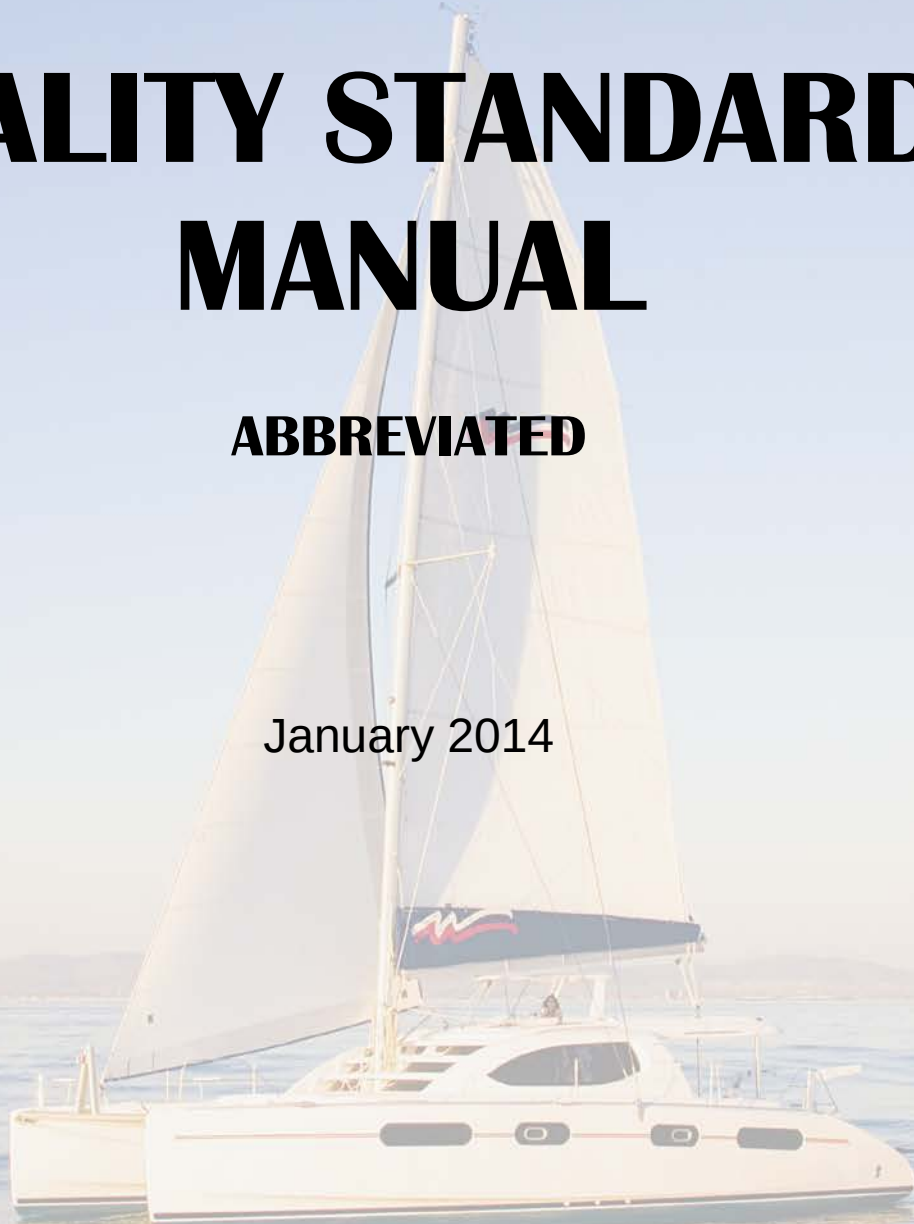


Robertson and Caine

QUALITY STANDARDS MANUAL

ABBREVIATED

January 2014



Quality Policy

Quality is not just high technology, skill, craftsmanship or hard work, it is an attitude, a refusal to compromise on the way in which these elements are brought together.

It is the policy of Robertson and Caine to provide products & services, which meet the needs and expectations of our customers, and to strive for continuous improvements in quality and productivity.

The Management of Robertson and Caine are aware that this cannot be achieved by themselves and the Quality Department alone, therefore we subscribe to the concepts of Right First Time, Defect Prevention and that quality is everyone's responsibility throughout the company.

Our goal is to build on our position as a world class manufacturer of luxury yachts with the objective of exporting quality products throughout the world.

I am personally committed to the Quality Management System and assurance is hereby given that the company quality policy is understood, implemented and maintained at all levels of the organisation.

J B Robertson
Managing Director

MISSION STATEMENT

We at Robertson and Caine, pride ourselves in being an international leader in the manufacture of top quality sailing yachts.

Within our Company.

We commit to:

Promoting the well being of our fellow staff and their families.

Offering each employee, performance recognition and a program of personal and skills development.

To our Clients.

We commit to:

Providing uncompromising Quality and Service.

Attaining excellent standards in productivity, safety and design.

We value.

Honesty and Integrity in all our relationships and transactions.

Our Mission is:

To diversify and develop a range of Marine Products for the Global Market, in order to secure a stable and rewarding environment for all stakeholders.

Determination of Inspection Standards

Visible Areas

Category A

High Visibility or Operationally Critical

Areas in this category include surfaces which are in areas of frequent use, decorative finishes, working areas and corian tops, all doors, floorboards, woodwork, upholstery, mirrors, glass, stainless steel and anodized surfaces. Also included in this category are all Instruments, Mechanical and Electrical fixtures and fittings, pumps, all through skin fittings and equipment which could affect the safety or Operation of the boat. ***The aesthetics of mechanical or electrical equipment is determined by the position of item in the boat.***

All items in this category must be free from operational faults or defects. Gel Coat surface areas must be free from defects such as Cracks, Pin Holes, Porosity, Scratches, Dirt / Dust Marks, loss of Gloss, Damage or Mould Marks. All Woodwork must be free of defects such as Scratches, Splits, Bubbles, Bad varnish, Contamination, Glue Marks, Delamination & Poor Joints.

All noted defects must be repaired to acceptable Quality Standard as specified in the relevant Company Operating Procedures. Defective or non-functional equipment must be replaced and returned to the supplier for review.

The Operation and functional suitability of all equipment must be 100% tested during Assembly, Final Inspection, Launch, Commissioning and Sea Trials.

Inspection Records, Manufacturing, Launch & Commissioning Files will be maintained by the Quality Assurance Manager for a minimum of 10 years from the Handover date of the Yacht.

Determination of Inspection Standards

Visible Areas

Category B

Intermediate Visibility and Textured Surfaces

Areas in this category include Textured surfaces and Gel Coat in non critical areas including Locker vertical panels, Step side panels, forepeak, Inner surfaces of all Locker lids and Engine compartments, all wooden Floors, Sealers and Joints in areas not specified in Category A.

Minor defects such as Small dust marks in Gel Coat, small surface defects that do not affect the function or safety of the component, Minor areas of Pin Holes, Porosity & Scratches may be allowable in these areas Woodwork may also have minor blemishes, loss of gloss & Small glue marks.

All noted defects must be repaired to acceptable Quality Standards as specified in the relevant Company Operating Procedures. Defective or non-functional equipment must be replaced and returned to the supplier for review.

The Visual and Aesthetic appearance of all Surfaces in this category must be monitored during Assembly, Final Inspection, Launch, Commissioning and Sea Trials. Inspection Records, Manufacturing, Launch & Commissioning Files will be maintained by the Quality Assurance Manager for a minimum of 10 years from the Handover date of the Yacht.

Determination of Inspection Standards

Visible Areas

Category C

Low Visibility

Areas in this category include Floor and Wall joints, Inner surfaces of Cupboards, Shelves & Lockers, side panels within 500mm of the floor under sides of Tables and Shelves, wooden supports & locker reinforcements Floors, Sealers and Joints in areas not specified in Category A or B. The waterline area below the hull is to be covered by antifouling.

Small defects such as Small dust marks in Gel Coat, small surface defects that do not affect the function or safety of the component, small areas of Pin Holes, Porosity & Scratches may be allowable in these areas Woodwork may also have small blemishes, loss of gloss & Small glue marks.

All noted defects must be repaired to acceptable Quality Standards as specified in the relevant Company Operating Procedures. Defective or non-functional equipment must be replaced and returned to the supplier for review.

Records of all repairs conducted must be retained and repairs visually monitored during Assembly, Final Inspection, Launch, Commissioning and Sea Trials. Inspection Records, Manufacturing, Launch & Commissioning Files will be maintained by the Quality Assurance Manager for a minimum of 10 years from the Handover date of the Yacht.

Determination of Inspection Standards

Visible Areas

Category D

Not Visible Immediately

Areas in this category include Bilges and areas under floorboards, GRP surfaces which are covered by removable vinyl trim panels. The under surfaces of cupboards and shelves, Engine Compartments & Water Tank Storage Lockers. Metallic surfaces for example; the underneath of toe-rails are sealed and therefore exposed anodizing in this area is accepted.

The acceptability of any defects must be reviewed in line with the relevant Company Procedures and the Allowable Defects list applicable to this category. Any defects which fall outside of these standards must be repaired.

All noted defects must be repaired to acceptable Quality Standards as specified in the relevant Company Operating Procedures. Defective or non-functional equipment must be replaced and returned to the supplier for review.

The Visual and Functional appearance of all Surfaces in this category must be monitored during Assembly, Final Inspection, Launch, Commissioning and Sea Trials. Inspection Records, Manufacturing, Launch & Commissioning Files will be maintained by the Quality Assurance Manager for a minimum of 10 years from the Handover date of the Yacht.

NB: NOTE THAT ALL AREAS ARE INSPECTED UNDER NATURAL LIGHT AT A DISTANCE OF 1M OR MORE
(Natural light from sunrise to sunset)

Category A	
Gel Coat	Allowable Defect
Dust Marks	Maximum of 3 dust marks in any 200mm individual marked area no larger than 3mm
Scratches	Not acceptable without corrective action
Star Cracks	Not acceptable without corrective action
Thin Gel Coat	Not acceptable in high visibility areas. Other areas acceptable where repair would cause difference in surface finish. All efforts must be taken to avoid thin gelcoat.
Pin Holes	Not acceptable without corrective action
Porosity	Not acceptable without corrective action
Loss of Gloss	All areas must be uniformly buffed
Chipped or Damaged	Not acceptable without corrective action
Mould Marks	Not acceptable without corrective action
Epoxy on Gel Coat	Not acceptable without corrective action
Gel Coat Up stands	Not acceptable without corrective action
Hairs in Gel Coat	Only individual hairs maximum 10mm in length
Track Mark Damaged	Not acceptable without corrective action
Pin Holes in Track Mark	One areas no more than 5mm in diameter allowable in any 200mm
Bubbles	Not acceptable without corrective action
Stripes	No smudges, lines, thin gelcoat are acceptable along the length of the stripe.

Category A	
Joinery & Corian	Allowable Defect
Glue Marks	Glue marks are not acceptable without corrective action. Glue marks < 1mm are acceptable if contained within a joint.
Bubbles (Fish Eyes)	No bubbles are acceptable without corrective action.
Rough Finish	Not acceptable without corrective action
Thin Varnish	Not acceptable without corrective action
Joint cracked	Not acceptable without corrective action
Delamination (Veneer)	Not acceptable without corrective action
Scratches	Not acceptable without corrective action, unless <10mm in length) and not clearly visible
Damaged/ Blemishes	Indentation maximum 3mm diameter 0.5mm deep no damage to varnish allowed, 1 per item
Pencil Marks	Not acceptable without corrective action
Varnish Runs	Not acceptable without corrective action
Joint proud	Not acceptable without corrective action
Unsealed end grain	Not acceptable without corrective action
Knots	Wet knots which do not detract from the finish of the wood are allowable with a cross section < 2.5mm and length <20mm.
Colour Mismatching	Not acceptable. Fiddles and capping's must match the finished wood panels within the tolerance of the standards. Light and dark striations are not acceptable

Category A	
Windows	Allowable Defect
Bubbles/Fish Eyes/Pinholes	1mm - 1 per sq. M. Over 1mm - none allowed. Less than 1mm disregard unless clusters form, causing visual defects.
Scratches	Allowable 1 fine scratch 10mm long in a 400mm radius. Must pass fingernail test and be repairable.
Blemish/Defect in material	Not acceptable unless defect occurs within 25mm of border/ edge, is < 20mm in size and not clearly visible (insignificant).
Chip marks	Not acceptable unless within the bonding area on the inside
Waves/ Ripples/ Distortions	Not acceptable if distortions of objects beyond the window are visible when looking through the window at 90 deg up to 45 deg, (not at the surface).
Glazing	Glazing must be even and uniform with straight edges along its entire length.
Tint	Must be from the same batch as for the entire boat

Category A	
Metallic Finishes / Mirrors (Decorative)/Saloon Doors	Allowable Defect
Pinholes	Not acceptable if visible from > 1m
Scratches	Not acceptable if visible from > 1m
Welding/ Anodising Clamp marks	Not acceptable if visible from > 1m
Anodising Discolouration	Not acceptable if visible from > 1m. All areas must be fully covered by Anodising.
Voids/ Lumps	Not acceptable
Blemishes/ Dents	Not acceptable
Corrosion	Not acceptable

Category A	
Upholstery/ Vinyl Liners/ Shade Cloth/ Bimini Covers	Allowable Defect
Colour deviations	As per specification, must not deviate from Master Sample
Fading	Not acceptable
Skew edging/Uneven seams	Not acceptable
Missing/Loose stitches	Not acceptable
Glue marks/ Pen marks/ Tears	Not acceptable
Poor Fitment/Sagging/Wrinkles	Not acceptable, must fit as per specification

Category A	
Floorboards	Allowable Defect
Colour deviations	Not acceptable, must not deviate from Master Sample
Warp	Not acceptable. It must not rock and must be level
Poor Fitment	Not acceptable. An all round even and parallel gap of 1- 1.5mm is mandatory.
Dents	The maximum indentation of 3mm diameter and 1mm deep is allowed with no damage to the veneer. No more than 1 allowed in any area of 600 mm diameter
Chip marks	Not acceptable if > 1mm and visible from >1 m.
Squeaking	Not acceptable.

Category A	
Sealing/ Corking	Allowable Defect
Smears	Not acceptable
Discolouration	Not acceptable
Bubbles/ dents	Not acceptable
Dirt	Not acceptable
Uneven edges	Not acceptable
Excess sealer	Not acceptable
Tackiness	Not acceptable
Uneven thickness/ inconsistency	Not acceptable

Category A

Hardware/ Hinges/ Deck fitting	Allowable Defect
Burred screws	Not acceptable
Protruding counter sunk screws	Not acceptable. The screws must fit flush to the surrounds
Mismatching Screws	Not acceptable. All screws must be the same type
Sika Smears	Not acceptable on the outside of the hardware.
Non protruding Nyloc Nut & Bolt	Not acceptable. 3-5 threads protruding
Washer Overlap	Not acceptable
Loose Hinges	Not acceptable
Misaligned Equipment	Not acceptable

Please Note: Colour coding as per drawings.

	Category A
	Category B
	Category C
	Category D

Category B

Gel Coat	Allowable Defect
Dust Marks	Not acceptable. Maximum of 4 dust marks in any 200mm individual marks area no larger than 10mm
Scratches	Not acceptable. Fine scratches maximum 5mm long which pass the Finger nail test are acceptable. No more than 4 in any 200mm Diameter
Star Cracks	Not acceptable
Thin Gel Coat	Acceptable where repair would cause difference in surface finish. All efforts must be taken to avoid thin gelcoat.
Pin Holes	Small pin holes maximum 60mm in diameter in any 200mm area
Porosity	Not acceptable without corrective action
Loss of Gloss	All areas must be uniformly buffed
Chipped or Damaged	Not acceptable without corrective action
Mould Marks	Not acceptable without corrective action
Styrene Attack	Not acceptable without corrective action
Epoxy on Gel Coat	Not acceptable without corrective action
Gel Coat Up stands	Not acceptable without corrective action
Hairs in Gel Coat	Not acceptable. Only individual hairs maximum 20mm in length
Track Mark Damaged	Not acceptable without corrective action
Pin Holes in Track Mark	Two areas no more than 50mm in diameter allowable in any 500mm
Bubbles	Not acceptable without corrective action

Category B

Joinery/Corian	Allowable Defect
Glue Marks	Small Glue marks are acceptable if contained in the finish of the wood if < 3mm
Bubbles (Fish Eyes)	No bubbles bigger than 1,5mm in any 75mm area
Rough Finish	Not acceptable without corrective action
Thin Varnish	Not acceptable without corrective action
Joint cracked	Not acceptable without corrective action
Delamination	Not acceptable without corrective action
Scratched	Not acceptable without corrective action, unless <20mm in length) and not clearly visible
Damages/ Blemishes	Indentation maximum 5mm diameter 0.5mm deep no damage to varnish allowed, 1 per item
Pencil Marks	Not acceptable without corrective action
Varnish Runs	Not acceptable without corrective action
Joint proud	Not acceptable without corrective action
Unsealed end grain	Not acceptable without corrective action
Knots	Wet knots which do not detract from the finish of the wood are allowable with a cross section < 3.5mm and length <30mm. Dry knots not acceptable without corrective action.
Colour Mismatching	Not acceptable. Fiddles and capping's must match the finished wood panels within the tolerance of the standards. Light and dark striations are not acceptable

Category B	
Antifouling	Allowable Defect
Runs/ spills	Not acceptable
Protrusions	Not acceptable unless defect occurs within 100mm of border/ edge and < 1.5mm
Holes/ uneven surfaces	Not acceptable

Category B	
Hardware/ Hinges/ Deck fitting/Traction Plated	Allowable Defect
Sharp edging	Not acceptable.
Burred screws	Not acceptable.
Protruding counter sunk screws	Not acceptable. The screws must fit flush to the surrounds
Mismatching Screws	Not acceptable. All screws must be the same type
Sika Smears	Acceptable on the inner side of the bolts. Not acceptable on the outside of the bolt
Non protruding Nyloc Nut & Bolt	Not acceptable. 3-5 threads protruding
Washer Overlap	Not acceptable
Loose Hinges	Not acceptable
Misaligned Equipment	Not acceptable

Category B	
Metallic Finishes / Mirrors (Decorative)/Saloon Doors	Allowable Defect
Pinholes	Not acceptable
Scratches	Not acceptable. Fine scratches maximum 5mm long which pass the Finger nail test are acceptable. No more than 5 in any 200mm Diameter
Welding/ Anodising Clamp marks	Not acceptable
Anodising Discolouration	Not acceptable. All areas must be fully covered by Anodising.
Voids/ Lumps	Not acceptable
Blemishes/ Dents	Not acceptable
Corrosion	Not acceptable

Please Note: Colour coding as per drawings.

	Category A
	Category B
	Category C
	Category D

Category C

Gel Coat	Allowable Defect
Dust Marks	Maximum of 10 dust marks in any 500mm individual marks area no larger than 15mm
Scratches	Fine scratches maximum 25mm long are acceptable
Star Cracks	Not acceptable
Thin Gel Coat	Acceptable where repair would cause difference in surface finish.
Pin Holes	Medium pin holes maximum 100mm in diameter in any 500mm area
Porosity	Medium porosity maximum 100mm in diameter in any 500mm area
Loss of Gloss	All areas must be uniformly buffed
Chips/ Damages	A 3mm nick maximum 0.5mm deep is allowable no more than 3 per surface except for under the waterline
Mould Marks	Not acceptable without corrective action
Styrene Attack	Not acceptable without corrective action
Epoxy on Gel Coat	Not acceptable without corrective action
Gel Coat Upstands	Maximum 0.5mm in any viewed area
Hairs in Gel Coat	Only individual hairs maximum 30mm in length
Pin Holes in Track Mark	Two areas no more than 50mm in diameter allowable in any 500mm
Bubbles	Not acceptable without corrective action

Category C

Joinery	Allowable Defect
Glue Marks	Small glue marks are acceptable if contained in the finish of the wood
Bubbles (Fish Eyes)	No bubbles bigger than 1.5mm in any 75mm area
Rough Finish	Not acceptable without corrective action
Thin Varnish	Not acceptable without corrective action
Joint cracked	Not acceptable without corrective action
Delamination	Not acceptable without corrective action
Scratched	Scratches maximum 10mm long are acceptable
Damages/ Blemishes	Small Indentation are acceptable no cracks or splits are allowed
Pencil Marks	Not acceptable if > 5mm.
Varnish Runs	Not acceptable without corrective action if > 10mm
Joint proud	Acceptable
Unsealed end grain	Not acceptable without corrective action
Knots	Acceptable if stable. Dry knots not acceptable without corrective action
Colour Mismatching	Not acceptable. Fiddles and cappings must match the finished wood panels within the tolerance of the standards.

Category C

Hardware/ Hinges/ Deckfitting	Allowable Defect
Burred screws	Not acceptable.
Protruding counter sunk screws	Not acceptable. The screws must fit flush to the surrounds
Mismatching Screws	Not acceptable. All screws must be the same type
Sika Smeers	Not acceptable unless defect occurs within 100mm of border/ edge and < 1.5mm
Non protruding Nyloc Nut & Bolt	Not acceptable. 3-5 threads protruding
Washer Overlap	Not acceptable
Loose Hinges	Not acceptable
Misaligned Equipment	Not acceptable

Category C

Metallic Finishes / Saloon Doors	Allowable Defect
Pinholes	Not acceptable if visible from >1 m
Scratches	Fine scratches maximum 25mm. Anodising must be intact. No more than 5 in any 200mm
Welding/ Anodising Clamp marks	Not acceptable if visible from >1 m
Anodising Discolouration	Allowable if < 100mm in a 500mm area . All areas must be fully covered by Anodising.
Voids/ Lumps	Not acceptable
Blemishes/ Dents	Not acceptable. 1 Small dent <10mm across and < 1mm deep per 500mm area is acceptable. Anodising must be intact.
Corrosion	Not acceptable

Please Note: Colour coding as per drawings.

	Category A
	Category B
	Category C
	Category D

Category D

Gel Coat	Allowable Defect
Dust Marks	Allowable
Scratches	Allowable
Star Cracks	Not acceptable
Thin Gel Coat	Allowable
Pin Holes / Porosity	Allowable
Loss of Gloss	Allowable
Chipped or Damaged	A 10mm nick maximum 1.0mm deep is allowable no more than 3 per surface
Mould Marks	Allowable
Epoxy on Gel Coat	Allowable 1% of area viewed
Gel Coat Up stands	Allowable
Hairs in Gel Coat	Allowable
Bubbles	Allowable
Flow coat	Should be smooth, clean and free from burrs

Category D

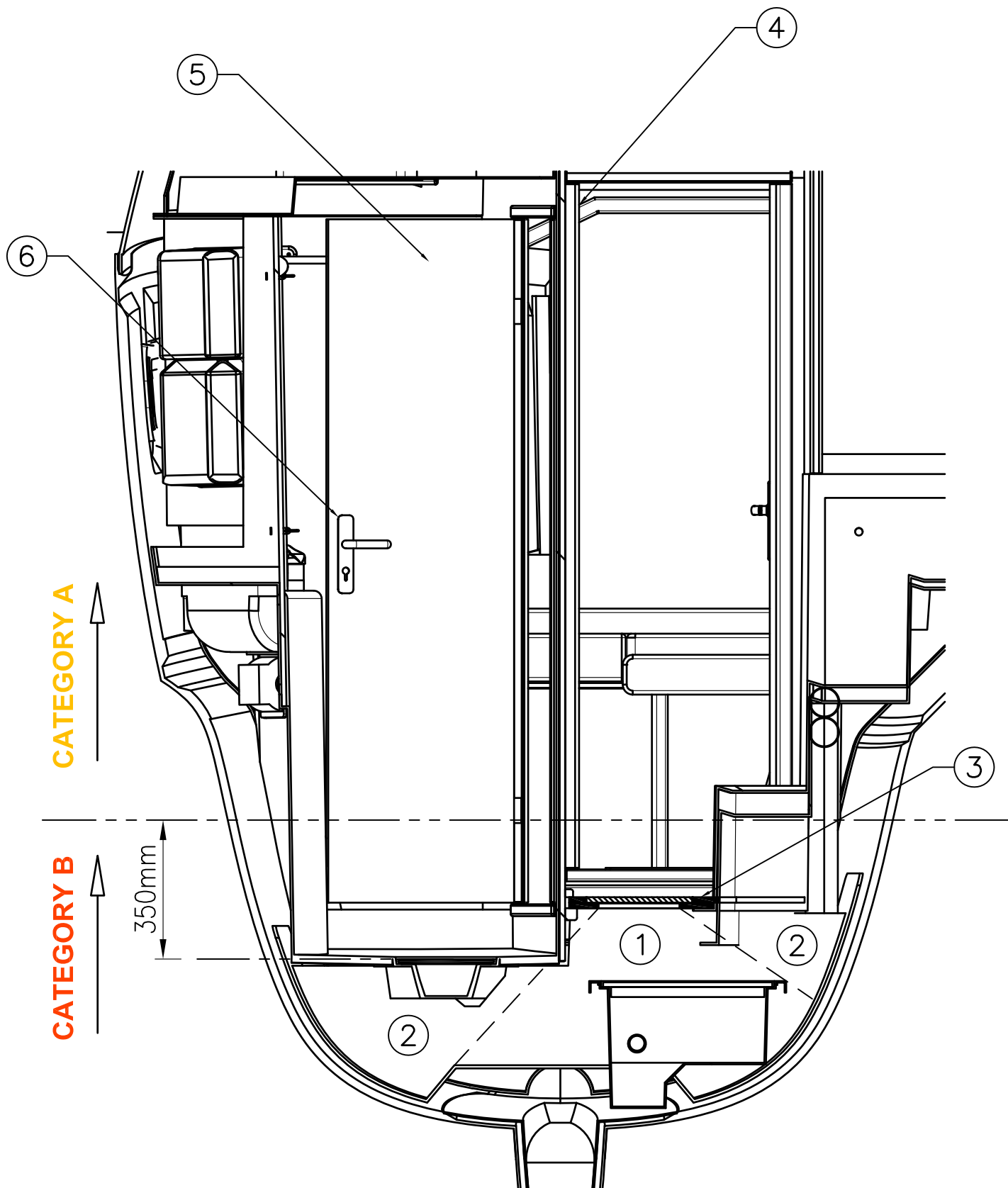
Joinery	Allowable Defect
Glue Marks	Acceptable
Bubbles (Fish Eyes)	Acceptable
Rough Finish	Acceptable
Thin Varnish	Not acceptable without corrective action
Joint cracked	Not acceptable without corrective action
Delamination	Acceptable
Scratched	Acceptable
Damaged/ Blemishes	Acceptable
Pencil Marks	Acceptable
Varnish Runs	Acceptable
Joint proud	Acceptable
Unsealed end grain	Must be sealed with either Flow coat or Varnish
Knots	Acceptable
Colour Mismatching	Acceptable

Category D

Hardware/ Hinges/ Deck fitting	Allowable Defect
Burred screws	Slight burring acceptable
Protruding counter sunk screws	Not acceptable. The screws must fit flush to the surrounds
Mismatching Screws	Not acceptable. All screws must be the same type
Sika Smears	Acceptable on the inner side of the bolts. Not acceptable on the outside of the bolt
Non protruding Nyloc Nut & Bolt	Not acceptable unless defect occurs within 100mm of border/ edge and < 1.5mm
Washer Overlap	Not acceptable
Loose Hinges	Not acceptable
Misaligned Equipment	Not acceptable

Category D

Metallic Finishes / Saloon Doors	Allowable Defect
Scratches	Fine scratches maximum 100mm allowable. Anodising must be intact.
Clamp marks / Discolouration	Acceptable
Blemishes/ Dents	Small dents < 1mm deep are acceptable as long as the function of the product is not jeopardised. Small areas without anodising <5mm is acceptable.
Corrosion	Not acceptable



CORRIDOR AREAS

NOTES:

1	UNDER SOLE FLOOR	C
2	UNDER SOLE FLOOR - OUT OF SIGHT	D
3	SOLE FLOOR	A
4	DOOR FRAME OR WALKTHROUGH CAPPING	A
5	DOOR	A
6	DOOR CANDLES & HINGES	A
AREA	DESCRIPTION:	CATEGORY:

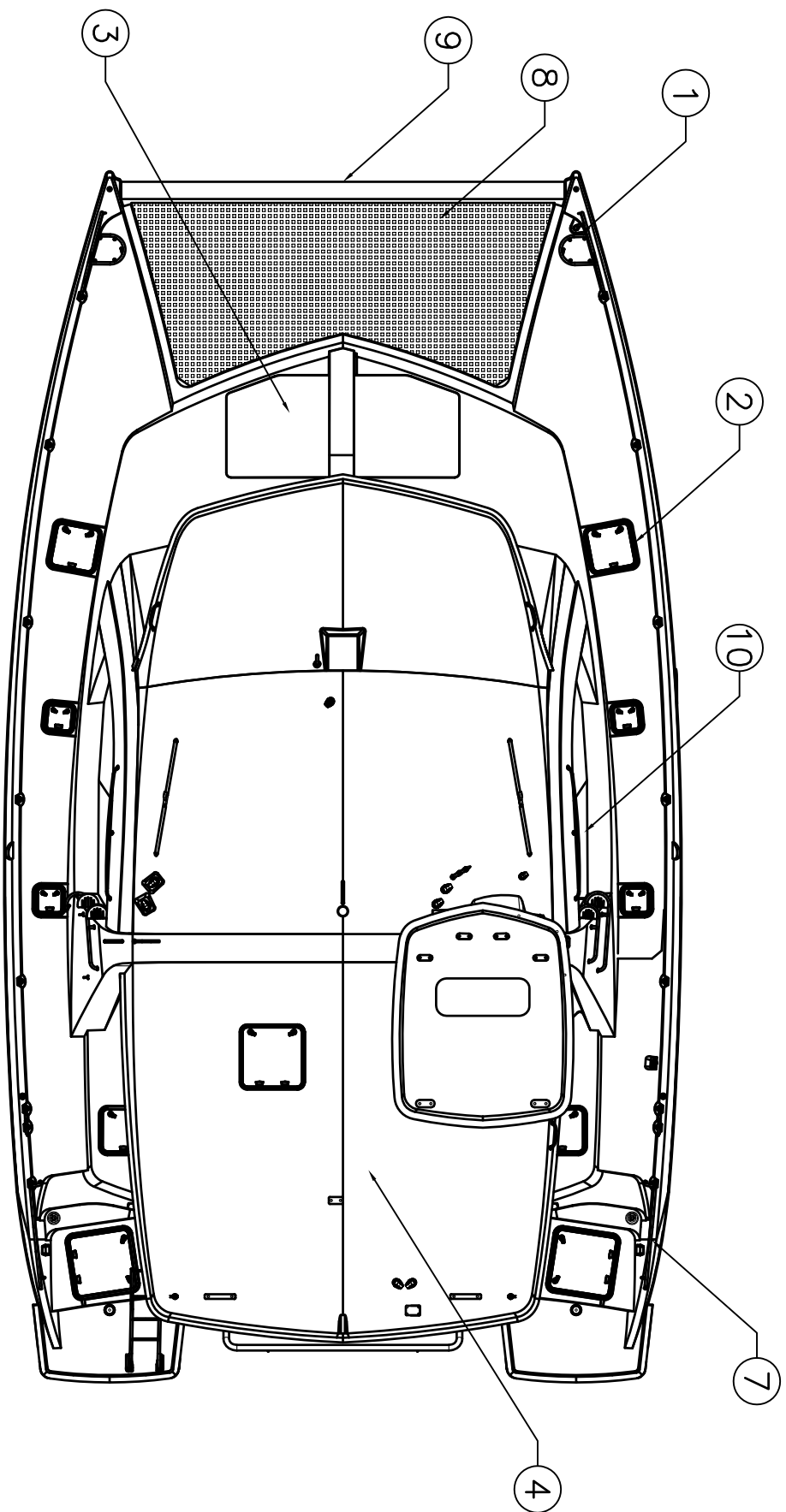


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TITLE: QUALITY MANUAL
CORRIDOR AREAS

SCALE: 1 : 10	DRAWN BY: D.ROUX DWG. NO: QA 001	APPROVED BY: DATE: 05/06/12
FILE NAME: QUALITY MANUAL DIAGRAMS	PAGE 06 of 06	



NOTES:

DECK AREAS

1	STAINLESS STEEL DECK FITTINGS	A	8	TRAMPOLINE	A
2	DECK HATCHES	A	9	CROSS BEAM	A
3	GRP LOCKER LIDS	A	10	WINDOWS	A
4	COCKPIT TABLE	A			
5	GELCOAT	A			
7	DECK GEAR	A			
6	GELCOAT TRACK MARK	A			
AREA	DESCRIPTION:	CATEGORY:	AREA	DESCRIPTION:	CATEGORY:

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DECK AREAS

SCALE:

1 : 20

DWG. NO: QA 001

APPROVED BY:

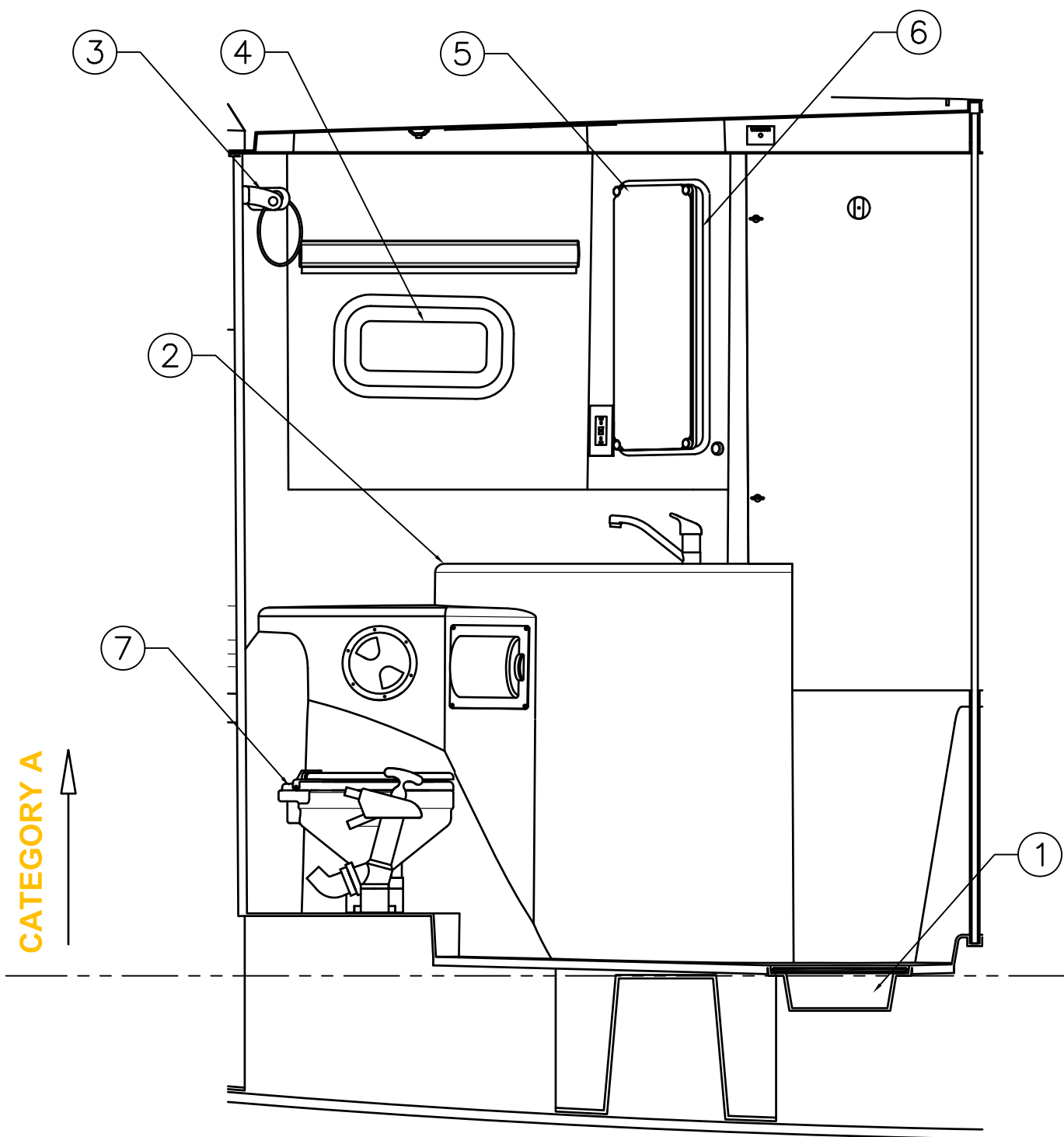
DATE: 05/08/12

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FILE NAME:

QUALITY MANUAL DWGQAMS

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HEADS AREAS

NOTES:

1	SHOWER SUMP	B
2	HEADS COUNTER TOP	A
3	FITTINGS - PLASTIC OR STAINLESS STEEL	A
4	PORTLIGHT	A
5	MIRROR	A
6	INSIDE VANITY LOCKER	A
7	TOILET	A
AREA	DESCRIPTION:	CATEGORY:



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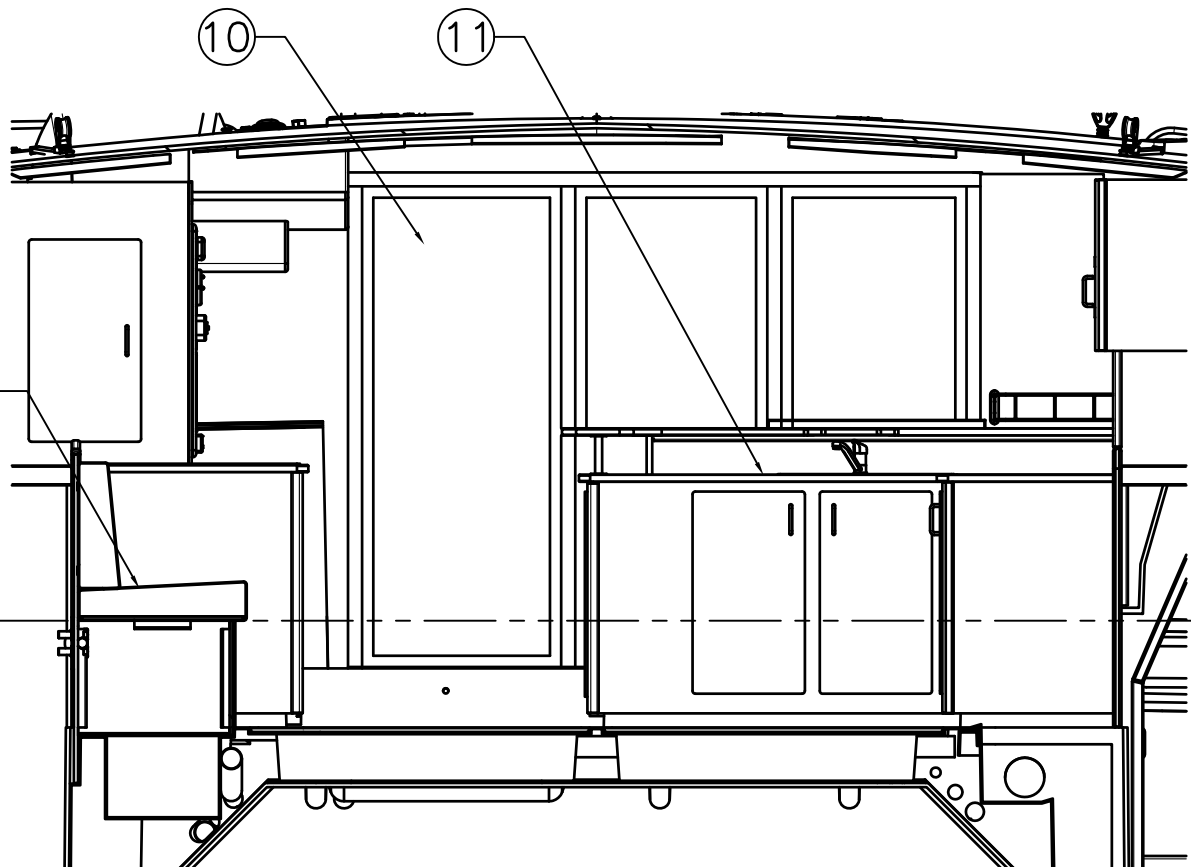
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TITLE:
QUALITY MANUAL
HEADS AREAS

SCALE: 1 : 10	DRAWN BY: D.ROUX DWG. NO: QA 001	APPROVED BY: DATE: 05/06/12
FILE NAME: QUALITY MANUAL DIAGRAMS	PAGE 04 of 06	

CATEGORY A

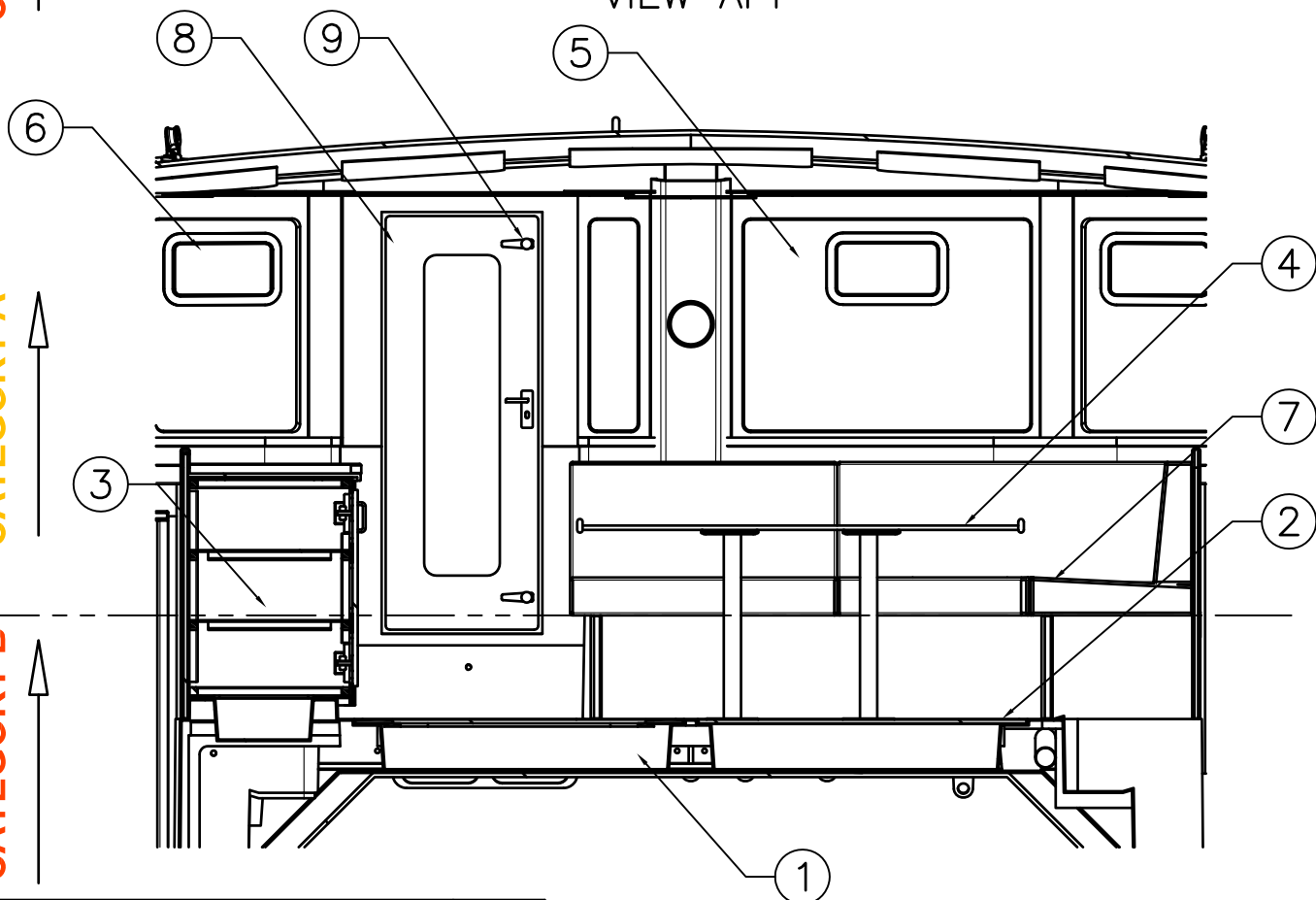
CATEGORY B



VIEW AFT

CATEGORY A

CATEGORY B



VIEW FORWARD

NOTES:

1	BELOW FLOORBOARDS	B
2	SOLE FLOOR	A
3	INSIDE LOCKERS	B
4	SALOON TABLE	A
5	WINDOWS	A
6	PORTLIGHTS	A
7	SETTEE CUSHIONS	A
8	FORWARD COCKPIT DOOR	A
9	STAINLESS STEEL FITTINGS	A
10	COMPANIONWAY SLIDING DOOR	A
11	GALLEY COUNTER TOP	A
AREA	DESCRIPTION:	CATEGORY:



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TITLE: QUALITY MANUAL SALOON AREAS		
SCALE: 1 : 20	DRAWN BY: D.ROUX DWG. NO: QA 001	APPROVED BY: DATE: 05/06/12
FILE NAME: QUALITY MANUAL DIAGRAMS	PAGE 03 of 06	

